Do Now · retrieval

Symmetric and asymmetric encryption

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

Which technique protects data against casual interception?

Q2

What does a network proxy do?

Q3

What is the difference between confidentiality and integrity?

Q4

Rising-edge D flip-flop Q=0, D=1 just before next edge. What happens to Q, and when?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Understand — Symmetric encryption uses the same secret key for encryption and decryption. The difficulty is sharing that key securely.

Core idea 2

Trace — Asymmetric encryption uses a public/private key pair. Public-key encryption for confidentiality uses a recipient's public key to encrypt and their private key to decrypt.

Core idea 3

Evaluate — Real secure connections typically use a hybrid protocol: authenticated asymmetric cryptography to establish secrets, then efficient symmetric encryption of application data.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

symmetric — Shared-secret encryption.

asymmetric — Public/private-key cryptography.

public key — Distributable key in a pair.

private key — Secret member of key pair.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Alice and Bob already share the secret key K. Which key does each use to protect and recover a file?

Follow the worked method

1. Predict first — Alice encrypts with the agreed secret K; an observer without K should not recover the plaintext.

2. Show method — Bob uses the same secret K to decrypt, so key secrecy and its secure distribution matter.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Predict first

Alice encrypts to Bob's public key without needing to learn Bob's private key.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Describe symmetric and asymmetric key arrangements. [6]

Question 2

Explain how a secure website can combine asymmetric and symmetric methods. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Evaluate symmetric-only encryption for a school communicating with 200 external partners. [6]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

A public key must be kept secret to use public-key encryption.

Use key vocabulary from memory

1. Explain symmetric in your own words.

2. Explain asymmetric in your own words.

3. Explain public key in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.